UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 3C – Engineering Anthropometry III

EIN 3241: Human Factors and Ergonomics
Spring 2026

February 2, 2026

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Admin Issues

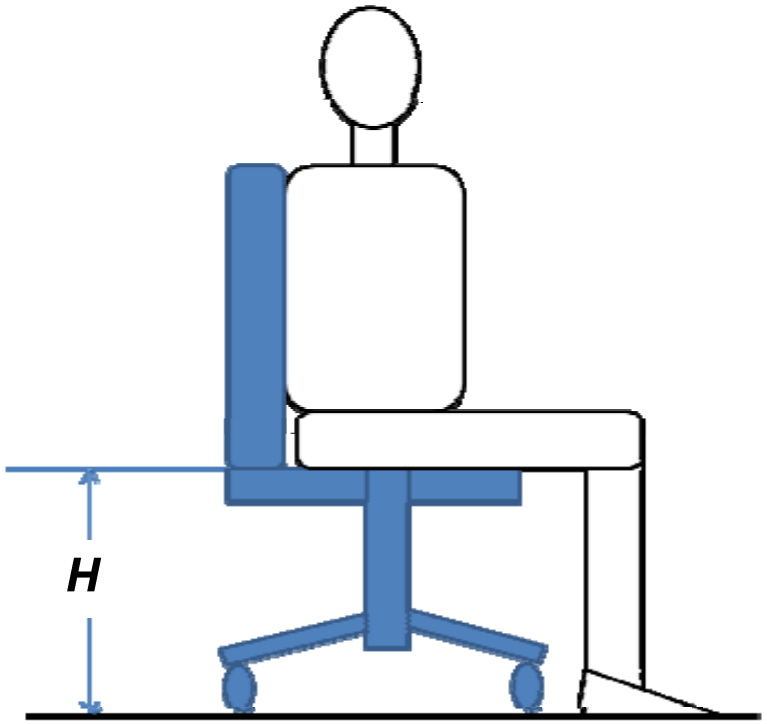
- Quiz 2
  - Next Class!

Recap…

- What is anthropometry
- Design applications
- Measurement devices and techniques
- Factors that affect anthropometric data
- Structural dimensions…
- Human Variability..
- Anthropometry Percentiles
- Anthropometric Design Principles
- ….

Anthropometry Problem 1

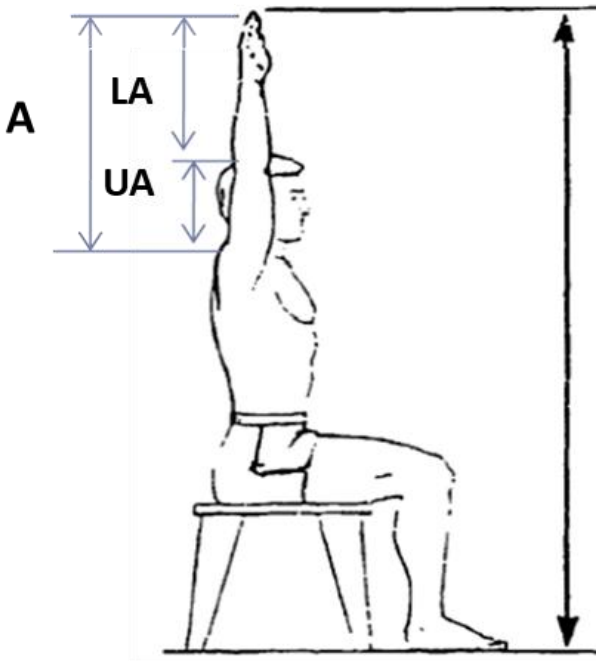
- As a Human Factors Specialist, your job is to determine the popliteal height, H of the sitting surface of a chair so that all the operators using this chair could sit on it comfortably. Comfortable posture is defined as lower leg perpendicular to the ground and upper leg parallel to the ground.



	Popliteal Height , H (cm)
Male	44.2 (2.8)
Female	39.8 (2.7)

Anthropometry Problem 2

- Calculate the 95th percentile shoulder-to-fingertip length. Given that the mean length of the lower arm (LA) link length (with the hand) is 442.9 mm with a standard deviation of 23.4 mm. The mean upper arm (UA) link length is 335.8 mm, and its standard deviation is 17.4 mm. The correlation between UA and LA is 0.4



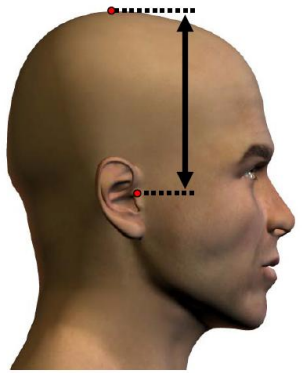
What if you had distributions of A and UA and want LA instead?

Anthropometry ANSUR Measures

- So far, we've focused on structural static anthropometry, measures like stature, eye height, and popliteal height, all taken in standardized, fixed postures.
- ANSUR (Anthropometric Survey of U.S. Army Personnel) is a comprehensive anthropometric survey originally conducted on U.S. military personnel.
  - It gives us over 90 standardized measurements per person, from the familiar (like stature and hip breadth) to more specialized ones (like bitragion chin arc or thigh circumference).
  - Static too!
- https://www.openlab.psu.edu/ansur2/

Tragion – Top of Head

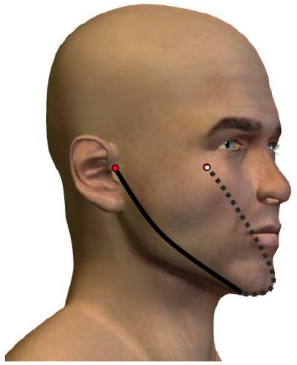
The vertical distance between the right tragion landmark and the horizontal plane tangent to the top of the head. For female participants with braids or cornrows, the measurement includes the styled hair.



Anthropometry ANSUR Measures

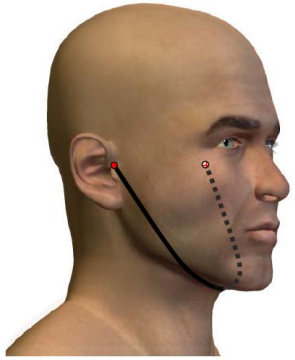
Bitragion Chin Arc

The surface distance between the right and left tragion landmarks across the chin landmark



Bitragion Submandibular Chin Arc

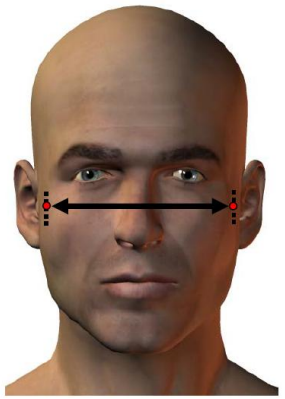
The surface distance between the right and left tragion landmarks across the submandibular landmark



Anthropometry ANSUR Measures

Bizygomatic Breadth

The maximum horizontal breadth of the face (between the zygomatic arches) at the left and right zygion landmarks.



Head Breadth

The maximum horizontal breadth of the head above the plane of attachment of the ears. For female participants with braids or cornrows, the measurement includes the styled hair.

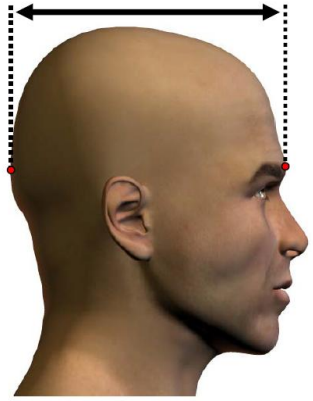
Anthropometry ANSUR Measures

Head Circumference

The maximum circumference of the head above the attachment of the ears is measured with a tape passing just above the ridges of the eyebrows and around the back of the head. For female participants with braids or cornrows, the measurement includes the styled hair.

Head Length

The distance from the glabella landmark to the opisthocranion landmark. For female participants with braids or cornrows, the measurement includes the styled hair.



Functional Data (Dynamic Anthropometry)

- Ergonomics isn't just about static dimensions (e.g., arm length), but about how far someone can comfortably and safely move their limbs or body.
  - For example: Can a worker reach a control without twisting awkwardly? Can they bend or rotate to see a monitor?
- Enter Functional Anthropometric Data
  - These are obtained with the body in various work postures.

TABLE 5.18 SQUATTING HEIGHT [CM (IN.)]

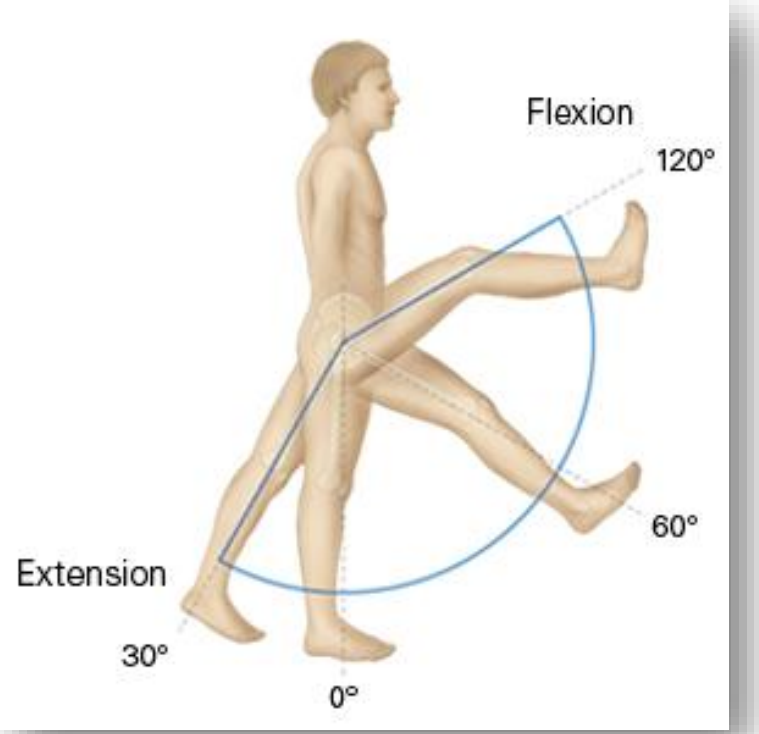
Subjects	Percentile		
	5th	50th	95th
U.S. Air Force personnel	103.6 (40.8)	110.7 (43.6)	119.4 (47)

Source: Adapted from Ref.[2].

Functional Data (Dynamic Anthropometry)

Joint Range of Motion (ROM)

- Why measure joint ROM data?
  - Define norms for population in extreme postures (reach limits)
  - Define extent of physical impairment after injury or disease
  - Workplace & tool design
  - Prevent awkward (extreme) postures
- What limits ROM?
  - For Active ROM
  - For Assisted Passive (or Volitional) ROM
    - Articular geometry (bone & cartilage)
    - Passive elasticity of connective tissues
    - Muscle shortening capacity
  - Normally: active ROM < passive ROM



Functional Anthropometers

Tool	Measures	Usage Context
Goniometers (manual or digital)	Joint angles	Clinical ROM assessment, ergonomics
Protractors / ELGON	Joint angles	Movement studies, occupational health
Slide Wire Goniometer	Joint displacement and angle	High-precision joint movement analysis
Strain Gauges (e.g., Penny & Giles)	Joint angle via electrical resistance	Wearables, high-resolution biomechanics
Optic Fibers / Data Gloves	Finger/hand joint motion	Wearable tech, VR, glove design, rehab
Inclinometers	Angular orientation of a body segment	Postural assessments, lifting studies
Flexometer	Spinal and torso flexion	Back movement assessments
Accelerometers	Acceleration of body segments	Dynamic motion capture, ergonomics of movement
Photogrammetry (2D, 3D)	Joint positions and movement	Motion analysis, posture evaluation
Video Motion Capture (2D, 3D)	Full-body motion in time	Sports, work task analysis, VR

Refresh Your Memory: Center of Mass (COM)

- The COM of an object is the average position of all small parts of that object weighted according to its total mass.
- Consider the COM to be located a distance x from the left edge of the segment:

$$M_x = \sum_{i=1}^{n} m_i x_i$$

$$x = \frac{1}{M} \sum_{i=1}^{n} m_i x_i$$

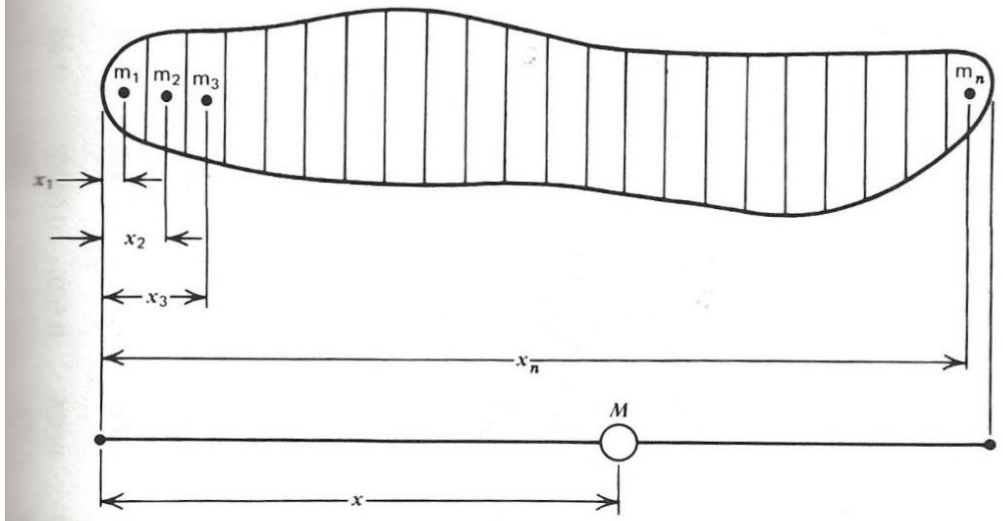


Figure 3.3 Location of the center of mass of a body segment relative to the distributed mass.

Assumptions: Segments are considered as Rigid Bodies and Joint Centers as Pin-Joint Centers.

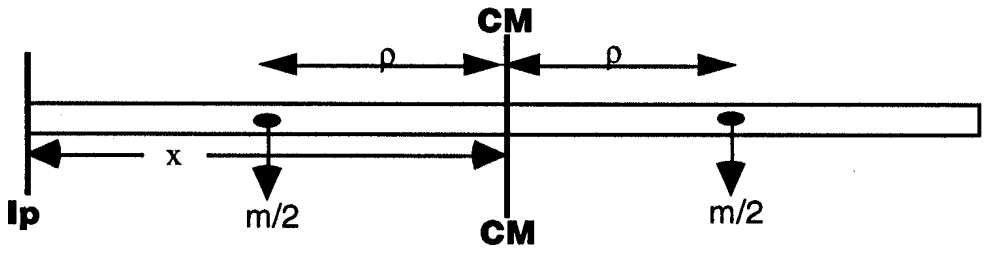
Refresh Your Memory: Mass, Moment of Inertia and Radius of Gyration

- **Mass and Inertia**: Mass is a measure of an object's resistance to changes in its motion. Inertia refers to an object's tendency to maintain its state of motion. The greater the mass, the greater the inertia.
- **Force**: A force is a push or pull that can cause an object to accelerate or decelerate.
- **Newton's Laws of Motion**: An object will remain at rest or move with a constant velocity unless acted upon by an external force. This equation also describes how forces cause changes in an object's motion
- **Newton's Laws of Equilibrium**: An object is in equilibrium when the sum of all forces acting on it is zero, and the sum of all moments acting on it is zero.
- **Moments of Inertia (MOI)**: MOI is an object's resistance to changes in its rotation.
- **Radius of Gyration**: It is the measurement (radial distance) of the distribution of atoms in a molecular structure with respect to either its center of mass or a given axis of rotation. Depends on size and shape, not on mass.

Mass, Moment of Inertia and Radius of Gyration

Parallel Axis Theorem

$$I_0 = \frac{m}{2}\rho^2 + \frac{m}{2}\rho^2 \ = m\rho^2$$



- The point ρ at which the point masses are equal distance from the center of mass is called the *Radius of Gyration*
- Since body segments rotate about joint centers rather than the CoM, we can now determine the MoI about the end of the segment

$$I_p = \frac{m}{2}\,(x-\rho)^2 + \frac{m}{2}\,(x+\rho)^2 = mx^2 + \ m\rho^2$$

$$I_p = mx^2 + I_0$$

where I_0 = moment of inertia about center of mass
x = distance between center of mass and center of rotation
m = mass of segment

Winter (1990)

- Dempster's Body Segment Parameters
- This is a gold-standard source in biomechanics and ergonomics for estimating:
  - Segment mass as a % of total body mass (P)

    E.g., $m_S = P_s \, m_{total}$
  - Center of mass (COM) location from the proximal end (R_proximal)
  - Radius of gyration (K_proximal) to calculate moment of inertia

Table 3.2 Dempster's Body Segment Parameters

Segment	Endpoints[a] (proximal to distal)	SEGMENTAL MASS/TOTAL MASS	CENTER OF MASS/SEGMENT LENGTH		RADIUS OF GYRATION/ SEGMENT LENGTH		
		$(P)^b$	$(R_{proximal})^c$	$(R_{distal})^c$	$(K_{cg})^d$	$(K_{proximal})^d$	$(K_{distal})^d$
Hand	Wrist center to knuckle II of third finger	0.0060	0.506	0.494	0.298	0.587	0.577
Forearm	Elbow to wrist center	0.0160	0.430	0.570	0.303	0.526	0.647
Upper arm	Glenohumeral joint to elbow center	0.0280	0.436	0.564	0.322	0.542	0.645
Forearm and hand	Elbow to wrist center	0.0220	0.682	0.318	0.468	0.827	0.565
Upper extremity	Glenohumeral joint to wrist center	0.0500	0.530	0.470	0.368	0.645	0.596
Foot	Ankle to ball of foot	0.0145	0.500	0.500	0.475	0.690	0.690
Leg	Knee to ankle center	0.0465	0.433	0.567	0.302	0.528	0.643
Thigh	Hip to knee center	0.1000	0.433	0.567	0.323	0.540	0.653
Lower extremity	Hip to ankle center	0.1610	0.447	0.553	0.326	0.560	0.650
Head	C7-T1 to ear canal	0.0810	1.000	0.000	0.495	1.116	0.495
Shoulder	Sternoclavicular joint to glenohumeral joint center	0.0158	0.712	0.288			
Thorax	C7-T1 to T12-L1	0.2160	0.820	0.180			
Abdomen	T12-L1 to L4-L5	0.1390	0.440	0.560			
Pelvis	L4-L5 to trochanter	0.1420	0.105	0.895			
Thorax and abdomen	C7-T1 to L4-L5	0.3550	0.630	0.370			
Abdomen and pelvis	T12-L1 to greater trochanter	0.2810	0.270	0.730			
Trunk	Greater trochanter to glenohumeral joint	0.4970	0.495	0.505	0.406	0.640	0.648
Head, arms, and trunk	Greater trochanter to glenohumeral joint	0.6780	0.626	0.374	0.496	0.798	0.621
Head, arms, and trunk	Greater trochanter to mid-rib	0.6780	1.142	−0.142	0.903	1.456	0.914

[a]Endpoints are defined in table 3.1.

[b]A segment's mass as a proportion of the total body mass.

[c]The distances from the proximal and distal ends of the segment to the segment's center of gravity as proportions of the segment's length.

[d]The radii of gyration about the center of gravity, proximal and distal ends of the segment to the segment's center of gravity as proportions of the segment's length.

Adapted from D.A. Winter 1990.

Mass, Moment of Inertia and Radius of Gyration

- Moment of Inertia (MOI) (Using the Table)

$I_0 = m\rho^2$ → MOI with respect to the Center of Gravity/Mass (K_{cg})

$I_p = m\rho^2$ → MOI with respect to the proximal end ($K_{proximal}$)

$I_d = m\rho^2$ → MOI with respect to the distal end (K_{distal})

- Moment of Inertia (MOI) (Using the parallel axis theorem and Table)

$I_p = mx^2 + I_0$ → x = distance between CoM and proximal end, ($R_{proximal}$)

$I_p = mx^2 + I_0$ → x = distance between CoM and distal end, (R_{distal})

where I_0 = moment of inertia about center of mass
x = distance between center of mass and center of rotation
m = mass of segment

MOI Example

- Calculate the center of mass of the leg and the moment of inertia about its proximal end. The body weight is 80 Kg. The actual leg (shank) length is 0.435 m.

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UF

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LEADING THE CHARGE, CHARGING AHEAD